House Price Prediction and Classification

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1 Introduction

This assignment applies a *machine learning* workflow to a housing dataset containing information about 1,208 houses and 19 variables. The dataset contains deliberate data-quality problems, including missing values, inconsistent categorical spellings, capitalisation differences, malformed numerical values, and duplicate property records.

Two machine learning tasks are performed:

Firstly, a **linear regression** model is trained to predict the continuous target variable `sale_price_eur`. This is a regression problem because the target is a numerical value representing the sale price of a house.

Secondly, a **logistic regression** model is trained to predict the binary target `high_price`. According to the assignment, `high_price` is equal to 1 when the sale price is at least EUR 500,000 and 0 otherwise. This is a classification problem.

The analysis follows the standard ML workflow: problem definition, data inspection and understanding, cleaning the raw data, exploratory data analysis, preprocessing, train/test splitting, model training, evaluation, and discussion of possible improvements.

2 Machine Learning Workflow

A machine learning workflow consists of several connected stages.

2.1 Defining the problem

The first task is to determine what should be predicted and which variables can be used as inputs.

For the linear regression problem,

$$X = \text{house characteristics} \quad \text{and} \quad y = \texttt{sale_price_eur}.$$

The model therefore predicts a continuous numerical value.

For the logistic regression problem,

$$X = \text{house characteristics} \quad \text{and} \quad y = \texttt{high_price}.$$

The target is defined as

$$\texttt{high_price} = \begin{cases} 1, & \text{if sale price} \geq EUR\,500,000, \\ 0, & \text{otherwise.} \end{cases}$$

Thus, linear regression is used for regression and logistic regression is used for binary classification.

2.2 Understanding the data

The original dataset contains 1,208 observations and 19 columns. The variables are `house_id`, `city`, `neighborhood`, `property_type`, `bedrooms`, `bathrooms`, `living_area_sqm`, `lot_size_sqm`, `year_built`, `condition`, `parking_spaces`, `distance_city_center_km`, `school_rating_1_10`, `crime_rate_1_10`, `has_garden`, `has_balcony`, `energy_rating`, `sale_price_eur` and `high_price`, and some are numerical while others and categorical.

An important data-quality issue is that some variables that are logically numerical are initially stored as text. For example, `bathrooms` contains values such as "1 baths", while `lot_size_sqm` contains values such as "90m2". These values must be cleaned before they can be used as numerical variables.

2.3 Data preprocessing

Preprocessing is necessary because machine learning algorithms require consistent numerical and categorical representations.

2.3.1 Missing values

The dataset contains missing values in several variables. In addition, the question mark character "?" is used to represent missing or invalid values. The question mark was therefore converted to a missing value (`NaN`).

For numerical variables, missing values were handled by median imputation (selected because it is less affected by extreme values than mean imputation), while the missing values in categorical variables were dealt with using the most-frequent category.

2.3.2 Incorrect numerical formats

Several numerical variables contained malformed text.

For example,

```
1 baths
90m2
```

were converted to

```
1
90
```

before conversion to numerical data types.

Invalid numerical values were converted to missing values and handled through the imputation pipeline.

2.3.3 Categorical inconsistencies

The dataset intentionally contains spelling and capitalisation errors. For example, different variants of the same city can occur, such as "Frankfrut" and "frankfurt".

These values were standardised so that different spellings referring to the same category were represented consistently. Similar corrections were performed for property types, condition, neighborhoods and energy ratings. Boolean variables such as `has_garden` and `has_balcony` were converted to binary numerical values: Yes = 1, No = 0.

2.3.4 Duplicate records

Duplicate property records were identified while ignoring `house_id`. This is important because the identifier itself does not describe the property. Eight duplicate property records were removed, leaving 1,200 observations for modelling.

2.3.5 Outliers and unreasonable values

A histogram and *boxplot* of sale prices were used to inspect the distribution and identify unusually large or small observations. Extreme prices were not automatically deleted. A high-priced house can be a genuine observation rather than an error. Removing it without evidence could therefore remove useful information. Similarly, a lot size of zero was not automatically treated as missing because apartments can reasonably have no private lot.

2.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed before model training to understand the structure of the data and investigate relationships between variables. The sale price distribution was visualised using a histogram and boxplot. A correlation matrix was also calculated for numerical variables.

The strongest numerical correlations with sale price were approximately:

Table 1: *Numerical correlations with sale price*

Variable	Correlation
Living area	0.730
Bedrooms	0.618
Lot size	0.604
Bathrooms	0.422
Parking spaces	0.415
Has garden	0.408
Year built	0.099
Has balcony	0.061
School rating	0.041

Living area has the strongest numerical correlation with sale price. This is intuitively reasonable because larger houses generally contain more usable residential space.

Bedrooms and lot size also show relatively strong positive relationships with price.

Categorical variables were also investigated using median prices by city, property type and condition.

The median prices were approximately:

Table 2: *Approximate median price by city*

City	Median price (EUR)
Munich	1,040,000
Hamburg	653,500
Frankfurt	600,500
Berlin	574,000
Cologne	569,000
Leipzig	370,000

The results indicate substantial differences between cities.

Property type also appears to be associated with price. The median prices were:

Table 3: *Approximate median price by property type*

Property type	Median price (EUR)
Semi-detached	948,000
Detached house	935,000
Terraced house	878,500
Apartment	393,000

These results suggest that both numerical and categorical variables contain useful information for predicting house prices.

Correlation was not, however, interpreted as proof of causation. A correlation measures the strength and direction of an association between numerical variables.

3 Part A: Linear Regression

3.1 Target and feature selection

The target for Part A:

```
sale_price_eur
```

The following variables were excluded:

- `house_id`: an identifier rather than a meaningful property characteristic

- `high_price`: derived from the sale price and therefore would introduce target leakage
- `high_price_created`: also derived from sale price
- `sale_price_eur`: the target itself

The remaining property characteristics were used as predictors.

3.2 Train/test split

The data was divided into 80% training data and 20% test data.

A fixed random state of 42 was used to make the experiment reproducible.

The test data was not used when fitting preprocessing transformations or the model. This helps prevent information from the test set from leaking into the training process.

3.3 Preprocessing

Numerical missing values were replaced with the median. Categorical missing values were replaced using the most-frequent category, and categorical variables were converted to numerical variables using *one-hot encoding*.

The `drop='first'` option was used so that one category from each categorical variable is used as the reference category. This reduces redundant dummy variables.

Scaling was not required for ordinary linear regression because the least-squares solution does not require numerical features to be on the same scale.

The preprocessing and model were implemented together in a *scikit-learn* pipeline. This ensures that the same transformations are applied consistently to training and test data.

3.4 Linear regression model

The linear regression model has the general form

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_p x_p.$$

The coefficients describe the estimated relationship between the predictors and the target, while holding the other model variables constant.

3.5 Evaluation

Three metrics were used as required by the assignment.

3.5.1 Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

The MAE measures the average absolute difference between actual and predicted prices. The model achieved approximately $MAE \approx \text{EUR}\,100,874$. Therefore, the average absolute prediction error was approximately EUR 101,000.

3.5.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}$$

The RMSE was approximately $RMSE \approx \text{EUR}\,140,585$. Because $RMSE$ squares errors before averaging, large prediction errors receive greater weight than under MAE.

3.5.3 Coefficient of determination

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

The model achieved $R^2 \approx 0.8796$. This means that approximately 87.96% of the variation in sale price in the test set is explained by the fitted linear regression model.

3.6 Interpretation of important price features

The EDA and linear regression results indicate that the most important price-related characteristics include living area, number of bedrooms, lot size, bathrooms, parking spaces and the presence of a garden.

Living area has the strongest numerical correlation with sale price, with a correlation of approximately 0.73. Bedrooms and lot size also have strong positive correlations, of approximately 0.62 and 0.60, respectively. `city`, `property_type` and `condition` are also important because the median price differs substantially between their categories.

The regression coefficients provide an additional way to understand the fitted model. Positive coefficients indicate that increasing a numerical variable, or belonging to a particular category relative to its reference category, is associated with a higher predicted price. Negative coefficients indicate the opposite relationship.

Coefficient magnitudes should be interpreted carefully because different variables use different units and categorical variables are represented using dummy variables.

4 Part B: Logistic Regression

4.1 Target definition

The target variable is

```
high_price
```

and is defined as

$$high_price = \begin{cases} 1, & \text{if } sale_price_eur \geq EUR\,500,000, \\ 0, & \text{if } sale_price_eur < EUR\,500,000. \end{cases}$$

After cleaning and duplicate removal, there were approximately 439 observations in class 0 and 761 observations in class 1.

4.2 Avoiding data leakage

The most important modelling restriction in Part B is that `sale_price_eur` must not be used as an input feature. The target `high_price` is directly calculated from `sale_price_eur`. If the sale price were included as a predictor, the model would effectively be given the information needed to calculate the target (**data leakage**).

The `house_id` column was also excluded because it is simply an identifier.

Therefore, logistic regression uses only information that would reasonably be available as explanatory information rather than the answer itself.

4.3 Logistic regression

Logistic regression predicts the probability that an observation belongs to class 1.

The logistic function is

$$P(Y = 1|X) = \frac{1}{1 + e^{-z}},$$

where $z = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$.

The output is a probability between 0 and 1.

A default threshold of 0.5 is then used to classify observations into class 0 or class 1.

4.4 Preprocessing

The same general preprocessing strategy was used for logistic regression, except that categorical variables were one-hot encoded.

Unlike the linear regression model, numerical variables were standardised using *standard scaling*.

Standardisation transforms a variable approximately according to

$$z = \frac{x - \mu}{\sigma}.$$

Scaling is useful for logistic regression because the optimisation algorithm generally performs better when numerical features are on similar scales.

The train/test split was stratified so that the proportion of high-priced and non-high-priced houses remained similar in both sets.

4.5 Evaluation metrics

The logistic regression model was evaluated using *accuracy*, *precision*, *recall*, the *F1 score* and a *confusion matrix*.

4.5.1 Accuracy

Accuracy is defined as:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

It represents the proportion of all predictions that were correct.

The model achieved an accuracy of 0.9292. Therefore, approximately 92.92% of the test observations were classified correctly.

4.5.2 Precision

Precision is given by:

$$Precision = \frac{TP}{TP + FP}$$

The model had a precision of 0.9412, so about 94% of all the houses that the model predicted to be high-priced were actually so.

4.5.3 Recall

Recall is defined by:

$$Recall = \frac{TP}{TP + FN}$$

The model's *recall* was 0.9474.. Hence, the model identified approximately 94.7% of all genuinely high-priced houses.

4.5.4 F1 score

The $F1$ score is the harmonic mean of precision and recall. It is given by:

$$F1 = 2\frac{Precision \times Recall}{Precision + Recall}$$

The model achieved an $F1$ score of 0.9443. This indicates a strong balance between precision and recall.

4.6 Confusion matrix

The confusion matrix was

$$\begin{bmatrix} 79 & 9 \\ 8 & 144 \end{bmatrix}.$$

Using the convention that rows represent actual classes and columns represent predicted classes, this means:

- 79 houses were correctly classified as not high-priced.
- 9 houses were incorrectly classified as high-priced.
- 8 high-priced houses were incorrectly classified as not high-priced.
- 144 high-priced houses were correctly classified as high-priced.

The model therefore made relatively few classification errors.

4.7 Interpretation of logistic regression coefficients

Logistic regression coefficients describe changes in log-odds.

A positive coefficient indicates that an increase in the corresponding feature is associated with higher odds of belonging to the high-price class. A negative coefficient indicates lower odds.

The odds ratio can be calculated as

$$Odds\ ratio = e^{\beta}.$$

An odds ratio greater than one indicates increased odds, while an odds ratio below one indicates decreased odds.

Because numerical features were standardised, their coefficients can be interpreted approximately in relation to a one-standard-deviation change in the feature.

Features such as living area, bedrooms, lot size, bathrooms, parking, property type and city are expected to be important because these variables contain information strongly associated with house value.

5 Model Comparison

The two models solve different problems and therefore use different evaluation metrics.

Table 4: *Model performance*

Metric	Linear Regression	Logistic Regression
MAE	EUR 100,874	–
$RMSE$	EUR 140,585	–
R^2	0.8796	–
Accuracy	–	0.9292
Precision	–	0.9412
Recall	–	0.9474
$F1$	–	0.9443

The linear regression model provides a relatively strong prediction of continuous house prices, explaining approximately 88% of the variance in the test set.

The logistic regression model also performs strongly, with accuracy of approximately 92.9% and $F1$ score of approximately 94.4%.

The models were not directly compared using these numerical scores because they solve different prediction problems.

6 How the Models Could Be Improved

We now suggest some improvements that could be investigated.

6.1 Feature engineering

Additional useful features could be created.

For example,

$$house_age = current_year - year_built$$

could be more directly interpretable than "year built".

For the regression model, price per square metre could also be investigated as an analytical variable, although it should not be created using the target when used as a predictor.

Interactions could also be considered. For example, the effect of living area might differ between apartments and detached houses.

6.2 Non-linear models

Linear regression assumes a linear relationship between the predictors and the target.

House prices may have non-linear relationships with variables such as distance to the city centre, living area and age.

Tree-based methods such as Random Forest, Gradient Boosting or Random Forest classification could therefore be investigated.

6.3 Feature selection

Some variables may contain relatively little predictive information. Feature selection could be used to investigate whether removing irrelevant or noisy features improves generalisation.

6.4 Cross-validation

The current evaluation uses a single train/test split.

K-fold cross-validation could provide a more stable estimate of model performance by training and evaluating the model on several different splits.

6.5 Outlier investigation

Large prediction errors should be investigated to determine whether they correspond to unusual but legitimate houses or data-quality problems.

7 Limitations

Firstly, the dataset is relatively small, so the conclusions cannot be generalised to all locations in German. Secondly, the model evaluates associations rather than proving causal relationships; for example, the positive relationship between living area and price does not prove that increasing living area alone causes a particular increase in price. Thirdly, linear regression may not capture all non-linear relationships. Finally, the performance results depend on the particular train/test split. (Cross-validation would provide a more robust assessment.)

8 Conclusion

This assignment demonstrated a quite comprehensive complete machine learning workflow on a realistic housing dataset.

The dataset first required substantial preprocessing. Missing values, malformed numerical entries, inconsistent capitalisation, spelling errors and duplicate property records were identified and corrected.

Exploratory analysis showed that living area, bedrooms, lot size, bathrooms, parking and garden availability have relatively strong relationships with house price. City, property type and condition also show substantial differences in median prices.

For Part A, a linear regression model was trained to predict the continuous sale prices while for Part B, logistic regression was used to classify houses as high-priced or not high-priced.

9 Appendix: Main Software Libraries

The analysis was implemented in Python using Google Colab.

The main libraries were:

- `pandas` for data manipulation;
- `numpy` for numerical operations;
- `matplotlib` for visualisation; and
- `scikit-learn` for preprocessing, modelling and evaluation.

The complete Python notebook contains the data-loading, cleaning, EDA, preprocessing, modelling and evaluation steps described in this report.